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Reduction of Tankage Construction Time Using Six Sigma Approach

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Abstract

This paper focuses on Six Sigma (DMAIC-Define, Measure, Analyze, Improve and Control) approach at a Project Management. DMAIC is an improvement-focused Six Sigma methodology. This methodology helps Project team to reduce the Tankage construction time with effective cost savings. It is an initiative for continuous improvement by the Hindustan Petroleum Corporation Limited. In the first phase of the DMAIC process, the aim of the study was defined in a SMART way (Specific Measurable Achievable Realistic and Time bound) to reduce the Tankage construction time since completion of POL Terminal mainly depends on the completion of Tankages within the Contractual Delivery Date. During the Measure phase, Tankage construction time was derived through a PERT analysis using statistical tools such as Process Mapping, SIPOC study and Brainstorming, then the Critical path of Tankage construction was identified through the Critical path method. In the Analyze phase, the Critical path was studied thoroughly using Pareto analysis, then possibility and feasibility of Reducing of the top three high lead time activities in the Critical path time was derived through Cost-Benefit analysis. Cost-effective solutions of Critical path activities were derived in the Improve phase. During the Control phase, effective control measures had been taken through PFMEA (Potential Failure Mode and Effect Analysis) 5S, Gemba and Poka-yoke. Project results evidenced that lead-time of Tankage construction was reduced by 32 days/tank achieving a 25% reduction in overall construction time. By implementing DMAIC approach in Project management, the overall target of Minimizing Lead times with cost effective methods can be materialized.

Keywords: Lead time, Six-Sigma, DMAIC, SMART, Pareto Analysis, PERT, Brainstorming, Process Mapping, SIPOC, Cost-Benefit Analysis, Critical path, Jack Method, PFMEA, 5S, Gemba, Poka-Yoke

Introduction

Dharmapuri Terminal holds strategic importance for HPCL as it is the first pipeline-fed terminal in India-Tamil Nadu. The project was executed during the first wave of the COVID-19 pandemic, a period marked by severe restrictions on the movement of manpower and materials. Despite these challenges, HPCL remained committed to completing the project within the planned timeline. The construction of the POL terminal involved several critical activities, with tankage construction being the most significant, as it largely dictates

the overall completion timeline. Historically, tankage construction tends to exceed scheduled deadlines. Recognizing this, HPCL proactively aimed not just to meet the Contractual Due Date (CDD) but to reduce the construction time itself.

To achieve this, we adopted the Six Sigma methodology, a powerful quality improvement tool extensively used in the automotive industry to minimize defects and enhance performance. In this project, we implemented the Six Sigma approach within project management to optimize outcomes with minimal resources. Among the two primary methodologies DMADV (used in design) and DMAIC (used across various applications), we focused on the DMAIC approach to systematically improve the tankage construction process and achieve timely project completion. By integrating DMAIC into project management, engineers could streamline resource use, identify and correct process bottlenecks, and embed controls to sustain improvements thereby enhancing both efficiency and schedule adherence.

DMAIC Approach in Project Management

The DMAIC process begins with the Define phase, which formally launches the project by building a shared understanding of the problem, project scope, objectives, and stakeholder roles. During this phase, the team crafts a Project Charter, maps the high-level process using SIPOC or equivalent tools, clarifies how Voice of Customer inputs translate into Critical-To-Quality (CTQ) metrics, and establishes SMART goals. A kickoff meeting brings stakeholders together to confirm responsibilities, timelines, and governance, ensuring the initiative starts with clarity and strong alignment.

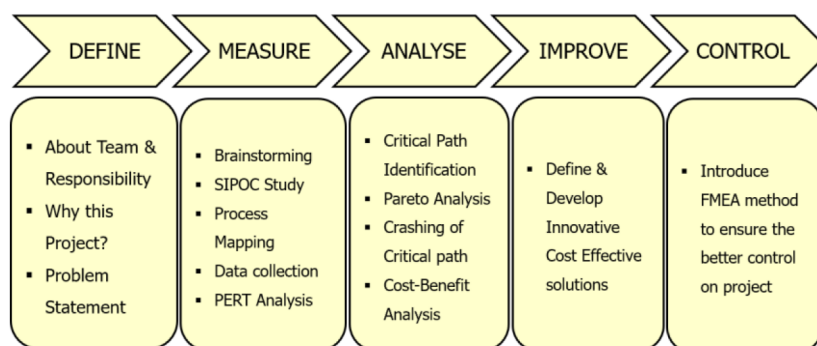


Figure 1—DMAIC Chart

In the Measure phase, the project team collects data to establish a clear baseline of the current process performance. Data gathering methods can include brainstorming exercises, SIPOC and process mapping, as well as observations and reviews. This stage defines key performance indicators and ensures data validity through rigorous checks. In the Analyze phase, the goal is to uncover root causes behind identified issues. Techniques such as Pareto charts, PERT or regression analysis, cause-effect diagrams, and cost-benefit comparisons are applied to validate contributing factors and focus on the most impactful ones. Once root causes are confirmed, the Improve phase generates and selects creative, cost-effective solutions. These are piloted, refined, and rolled out with updated process maps, ensuring alignment with project objectives. Finally, the Control phase ensures sustainability—new processes are documented via SOPs and control plans, stakeholders are trained, and monitoring tools such as PFMEA and statistical control charts are used to maintain improvements. This approach creates a disciplined, data-driven method for lasting process optimization.

Define Phase

Define phase defines the main problem and initial targets were defined in a systematic way.

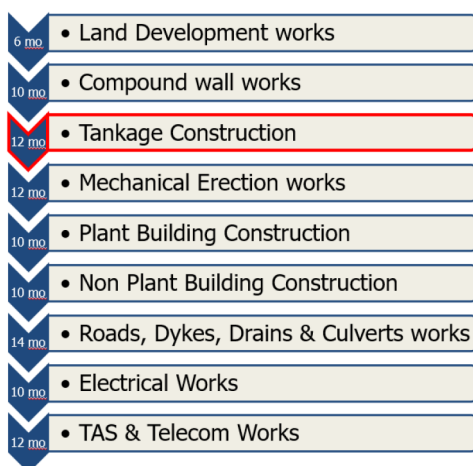


Figure 2—Major list of activities involved in the construction of Terminal.

Project Selection

Construction of POL Terminal includes Tankage works, Mechanical Works, Compound Wall & Land development works, TAS & Telecom works, Plant & Non-Plant Building works, Roads and Culverts works. Based on the earlier experiences and past data's, Tankage construction falls under the critical path and commissioning of POL terminal mainly depends on the completion of Tankages. So, Project team has decided to initiate the project on Reduction of Tankage Construction time.

22 Above Ground tanks and 11 underground tanks ([Table-1](#)) were planned at Dharmapuri Terminal. Among these 33 tanks, MS and HSD Tanks comprise 90% of total storage capacity and Tank weight in MT ([Figure-3](#) & [Figure-4](#)), hence focus shifted towards MS & HSD Tanks.

Table 1—No of Tanks constructed in Dharmapuri Terminal

Sl.No	Tank Service	No of Tanks
1	HSD Tanks	4
2	MS Tanks	4
3	SKO Tanks	4
4	Ethanol Tanks	2
5	Bio Diesel Tanks	2
6	Interface A Tank	1
7	Interface B Tank	1
8	Slop A Tank	1
9	Slop B Tank	1
10	UG tanks	11

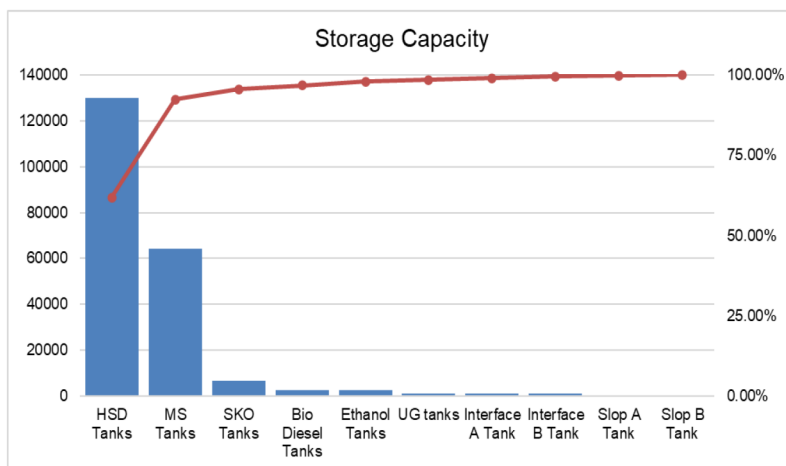


Figure 3—Pareto Analysis (Tankage total storage Capacity in KL)

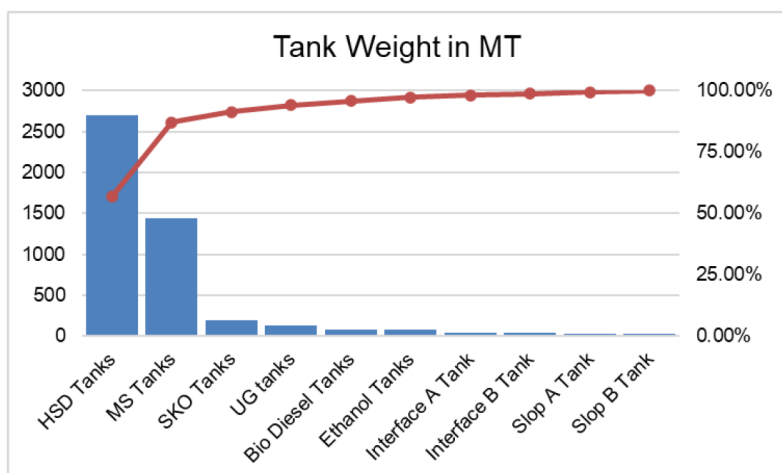


Figure 4—Pareto Analysis (Tankage Weight in MT)

A Project Charter (Figure-5) was prepared in a SMART (Specific Measurable Achievable Realistic and Time bound) way to achieve the Project goals in a cost effective way. The Objective of the Project has been taken as Reduction of Tankage Construction time by 10%. A Cross Functional Team (CFT) was formed and comprised of M/S.HPCL as Client, M/S.MECON ltd as EPCM and M/S. Bridge & Roof India Limited as a Construction contractor. Initially 5W2H analysis was done with the CFT for a better understanding of project initiation. Key process variables and project milestones were identified with specific time frames in Define phase.

BIG Y (OBJECTIVE)		Reduction of Tankage Construction time <u>Target: 10% reduction</u>		▪ Project Team: CLIENT: M/S.HPCL CONSULTANT: M/S.MECON Ltd CONTRACTOR:M/S.B&R India Ltd					
5W2H				Key Process Variable	D	M	A	I	C
What is the problem?	Tankage Construction time exceeds beyond Contractual Delivery time								
Why is it a problem?	Tankage Construction falls under critical path in Construction of POL Terminal			Tankage Construction time	Month-1	Month-1	Month-2	Month-4	Month-5
Where it was detected?	MS & HSD Tanks (Out of 22 Tanks MS & HSD Tanks comprises 90% storage capacity, Tonnage erection and Critical for commissioning...)								
Who detected it?	Cross Functional Team comprises of Client: M/S.HPCL-, EPMC-M/S.MECON Ltd & Vendor: M/S Bridge & Roof India Ltd								
When it was detected?	During Project 1 st Kick off meeting								
How it was detected?	During Project Kick off Meeting of Tankage Construction								
How many/much?	Mechanical Completion 169 days: Target: Reduction of 17 days/tank								

Figure 5—Project Charter

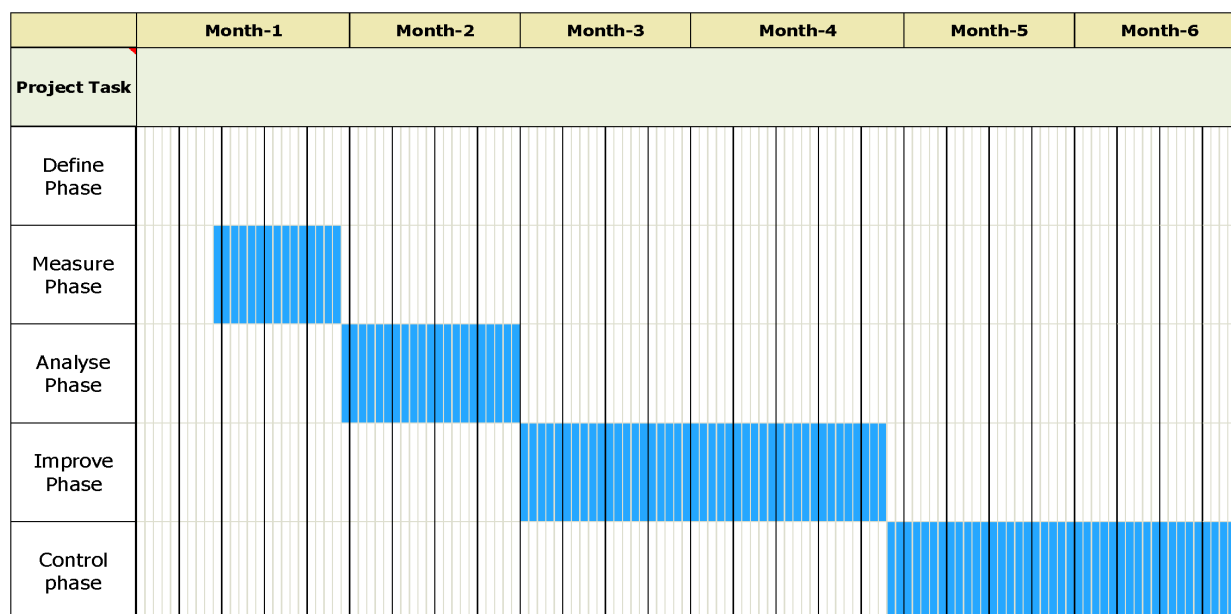


Figure 6—Project Plan (GANTT Chart)

Measure Phase

In Measure phase, Tankage construction time was derived through a PERT analysis using statistical tools such as Process Mapping, SIPOC study and Brainstorming.

PERT Network:

The Program Evaluation Review Technique, commonly known as PERT is used in project planning and analyses. It is commonly used to evaluate and estimate the time required to complete the project.

Step-1: Identifying the Activities and milestones

All major activities and related milestones were identified.

Step-2: Determining the sequence of activities

Activity sequence was identified with predecessors (Table-2)

Table 2—Sequence of Activities

Sl.No	Activity	Activity Description	Predecessor
1	A	Land Development	-
2	B	Excavation	A
3	C	Sand Filling	B
4	D	Bitumen Concrete	C
5	E	Plate fabrication	-
6	F	Bottom plate erection	D,E
7	G	Annular plate erection	D,E
8	H	Shell plate erection	G
9	I	Pontoon plate erection	F,H
10	J	Deck plate erection	F,H
11	K	Deck Plate works	I,J
12	L	Shell appurtenances	H
13	M	Deck appurtenances	K
14	N	Hydro testing	L,M

Step-3: Process Mapping



Figure 7—Process Mapping

Step-4: Estimating the time required for each activity (Table-3)

Table 3—Time Estimation Table

Sl.No	Activity	Activity Description	Predecessor	To	Tp	Tm	Tc=(To+Tp+4Tm)/6
1	A	Land Development	-	3	3	3	3.00
2	B	Excavation	A	6	14	8	8.67
3	C	Sand Filling	B	15	25	18	18.67
4	D	Bitumen Concrete	C	3	3	3	3.00
5	E	Plate fabrication	-	24	36	30	30.00
6	F	Bottom plate erection	D,E	10	18	10	11.33
7	G	Annular plate erection	D,E	9	18	15	14.50
8	H	Shell plate erection	G	16	24	20	20.00
9	I	Pontoon plate erection	F,H	8	13	12	11.50
10	J	Deck plate erection	F,H	10	22	15	15.33
11	K	Deck plate works	I,J	40	48	46	45.33
12	L	Shell appurtunances	H	10	18	15	14.67
13	M	Deck appurtunances	K	8	14	10	10.33
14	N	Hydro testing	L,M	24	38	30	30.33

This model works with three-time estimates.

- Optimistic time – The time required to complete the activity if everything goes right.
- The most likely time- The time required to complete the activity if everything goes Normal/Average
- Pessimistic time - The time required to complete the activity if everything goes wrong

$$\text{Expected time} = (\text{Optimistic} + 4 \times \text{Most likely} + \text{Pessimistic}) / 6$$

Analyze Phase

The Analyze phase is particularly important and aims to analyse the process in detail. The Critical path was studied thoroughly using Pareto analysis. A Network Diagram was constructed to identify the Critical path of the Project and calculate the Float of each activity.

Network Diagram Construction

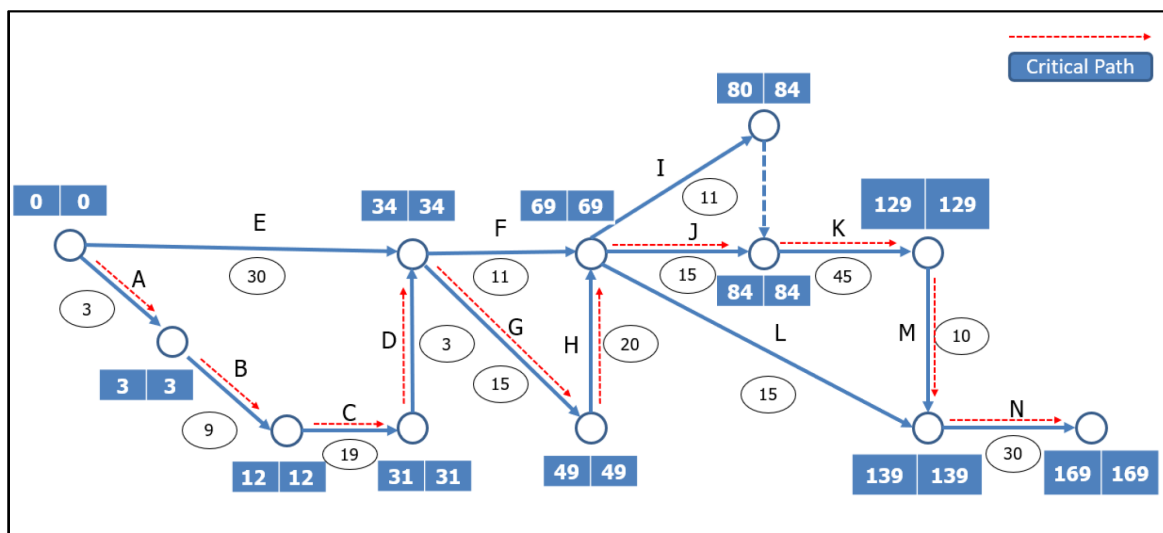


Figure 8—Network Diagram of Tankage Construction

Critical Path of the Project: A-B-C-D-G-H-J-K-M-N

Table 4—Activity Classification table

Activity	Activity Description	Classification	Te
A	Land Development	Critical Activity	3.00
B	Excavation	Critical Activity	8.67
C	Sand Filling	Critical Activity	18.67
D	Bitumen Concrete	Critical Activity	3.00
E	Plate fabrication	Non Critical Activity	30.00
F	Bottom plate erection	Non Critical Activity	11.33
G	Annular plate erection	Critical Activity	14.50
H	Shell plate erection	Critical Activity	20.00
I	Pontoon plate erection	Non Critical Activity	11.50
J	Deck plate erection	Critical Activity	15.33
K	Deck plate works	Critical Activity	45.33
L	Shell appartunances	Non Critical Activity	14.67
M	Deck appartunances	Critical Activity	10.33
N	Hydro testing	Critical Activity	30.33

Critical Path Analysis

Pareto Analysis of Critical Path:

The Pareto principle states that 80% of a project's benefit comes from 20% of the work. Also this technique is used to Identify and prioritize the problems and tasks.

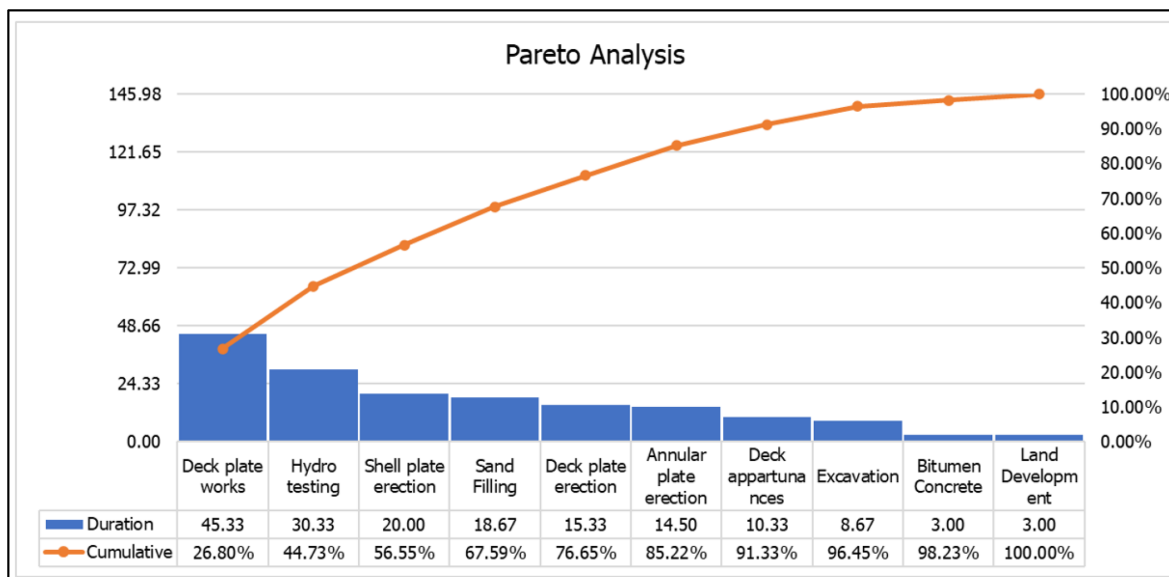


Figure 9—Pareto Analysis of Critical Path

Top 3 Activities in the critical path contribute 56.55% of construction time (Table-5).

Table 5—Pareto Analysis Table

SL.No	Activity Description	Te	Ind %	Cum %
1	Deck plate works	45.33	26.80%	26.80%
2	Hydro testing	30.33	17.93%	44.73%
3	Shell plate erection	20.00	11.82%	56.55%
4	Sand Filling	18.67	11.03%	67.59%
5	Deck plate erection	15.33	9.06%	76.65%
6	Annular plate erection	14.50	8.57%	85.22%
7	Deck appartunances	10.33	6.11%	91.33%
8	Excavation	8.67	5.12%	96.45%
9	Bitumen Concrete	3.00	1.77%	98.23%
10	Land Development	3.00	1.77%	100.00%
	Grand Total	169.17	100.00%	

Top 3 Activities in Critical Path

1. Deck plate works
2. Hydro testing
3. Shell plate erection

Cost Benefit Analysis

Cost benefit analysis was performed for the top 3 contributors in the Critical path.

Improvement ideas were identified by the Cross Functional Team with cost effective way

Sl.No	Activity Description	Base line	Target	Feasible (Y/N)	Improvement Idea	Investment Required	Cost Benefits	Benefit/ Investment Ratio	Final Feasibility status
1	Deck plate works-Deck Plate lifting	23	12.67	Yes	Deck Lifting using Jack Method	₹ 5,50,500	₹ 1,20,85,359	22	Feasible
2	Deck Plate works-Rolling Ladder Erection	22.33	10	Yes	Pre Fabricated Rolling Ladder Erection	₹ 1,50,000	₹ 1,33,000	0.90	Feasible
3	Hydro Testing	30.33	20.33	Yes	Identifying and engaging different resources for availability of Water	NA	NA	NA	Feasible
4	Shell plate erection	20.00	10.00	Yes	Engaging the Optimum no of welders at every stage as per requirement	NA	NA	NA	Feasible

Figure 10—Cost Benefit Analysis

Improve Phase

Improve phase aims to identify potential solutions with cost effective way. Three major potential solutions were proposed by the Cross Functional team

1. Deck Lifting using Jack Method
2. Prefabricated Rolling Ladder erection
3. Engaging optimum no of welders at every stage as per the requirement

Deck Lifting using Jack Method

As a normal practice, initial lifting in Deck plate erection is done (about 1.5m) using Water for Sleeve pipe and other miscellaneous works at Deck plate. This existing process of lifting by water takes 15 to 20 days/tank. It requires an enormous amount of water (about 32,000 KL for 8 EFR tanks) and also it has More Non-Value Added (NVA) activities resulting in reduction of Labour efficiency and effectiveness.

Solution Approach. JACK method can be an Innovative solution for the given problem and Challenges. Jacking method is the advanced form of tank erection and Hydraulic Jacks are used to lift the Tank shell during the Tankage construction. The Proposed Idea is to use the Inverted Hydraulic Jack (Figure-9) to lift the Tank Deck about 1.5m.



Figure 11—Jack Arrangement

Sequence of Process.

- After completion of all mechanical works related to bottom plate, shell plate and single deck floating roof, make-up shell plate of first shell course are to be fitted and welded.
- Shell manholes to be kept open to facilitate the connection between the hydraulic hose from hose junction to power pack.
- Permanent sleeve pipe in Deck plate can be used for Deck lifting. Square bar is inserted into each Temporary leg pipe which is connected into each Permanent sleeve pipe
- A Temporary leg must be inserted into each permanent sleeve pipe
- All hydraulic jack are inverted and fixed at the top plate.
- All individual hoses of jack to be connected at hose junction which is connected to power pack
- When Hydraulic pressure was applied at Jack piston it will pull the square bar which is inside the sleeve.
- Floating roof will be lifted on each stroke.
- After lifting of floating roof at the desired level the temporary legs pull out one by one and a permanent leg must be inserted into each sleeve.
- This innovative solution helps to achieve the Deck plate lifting by 5 days resulting in savings of 22 days per tank.



Figure 12—Sequence of Process

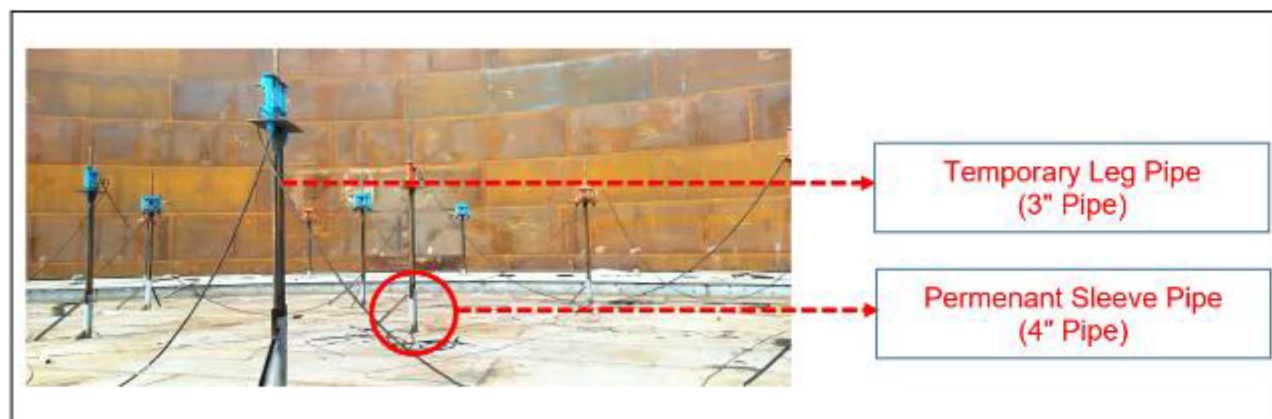


Figure 13—Temporary Leg Pipe Arrangement

Pre-Fabricated Rolling Ladder erection

Rolling ladder fabrication will start after completion of Deck plate lifting. This process takes 15 to 20 days/tank. Project Team has decided to change the sequential activities into parallel activities by Outsourcing the Rolling ladder fabrication. Pre-Fabricated Rolling ladder (FIGURE-12) is designed in such a way that there is no change in cost difference for fabrication and only the transportation cost is added to it.

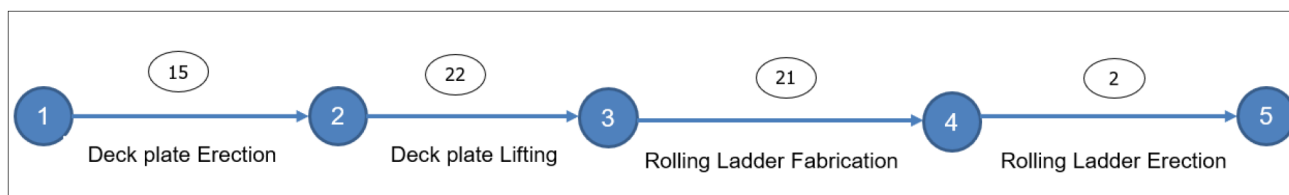


Figure 14—Network Diagram-Before Improvement

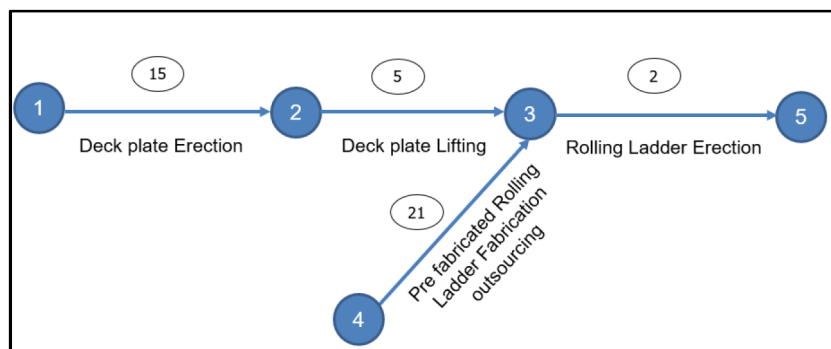


Figure 15—Network Diagram-After Improvement



Figure 16—Pre-Fabricated Rolling Ladder

Engaging Optimum no. of Welders as per the Requirement

- Detailed analysis and study were done along with Cross Functional Team for Calculation of Quantum of Work, Welders and Other Machineries requirement.
- In the beginning of the project, quantum of work (TABLE-7) was calculated and shell plate, annular plate, bottom plate and deck plate welding length and welding positions were calculated and accordingly manpower and machinery (TABLE-8) were mobilized by the contractor.
- Close monitoring was done by the Client and Consultant for Stage wise Inspection. All Inspection delay was captured digitally and routed to project head for necessary actions and leading to zero inspection delay for 84 days out of 127 days.
- Labour performance Incentive scheme and Overtime was directly monitored by the Client, Consultant and Contractor

Table 6—Welding Dimensions Details

Welding Length Dimensions		
Type of Joints	Tank	Length in m
Horizontal Joints	MS Tank	116.80
Vertical Joints	MS Tank	33.00
Horizontal Joints	HSD Tank	163.36
Vertical Joints	HSD Tank	47.64
Total Shell Length	MS Tank	298.00
Total Shell Length	HSD Tank	422.00
Annular Plate Welding	MS Tank	23.66
bottom Plate	MS Tank	525.00
Annular Plate Welding	HSD Tank	35.78
Bottom Plate Welding	HSD Tank	800.00
Deck Plate Welding	MS Tank	400.00
Pontoon Plate Welding	MS Tank	600.00

Table 7—No of Welders details

Optimum no of welders & Machineries		
Category	Tank	No of Welders Required
2G/3G Position	MS Tank	28
3F Position	MS Tank	12
Fabrication	MS Tank	6
2G/3G Position	HSD Tank	42
3F Position	HSD Tank	18
Fabrication	HSD Tank	9
Total No of Welders		115

Table 8—Machinery Requirement

Machinery Requirement	Tank	No of M/C Required
No of Welding M/C Required	MS Tank	46
No of Welding M/C Required	HSD Tank	69
No of 125 KVA DG	MS Tank	4
No of 63 KVA DG	MS Tank	1

Control Phase

In Control phase effective control measures were taken through PFMEA (Potential Failure Mode and Effect Analysis). To have a proper control on the project, the PFMEA tool has been used to identify the potential failure modes and address it before starting the project

PFMEA Analysis

A Potential Failure Mode Effects Analysis (PFMEA) is a structured analytical tool used by a cross-functional team to identify and evaluate the potential failures of a process to have a better control on the process and activities. PFMEA evaluates each process step and score assignment shall be carried out as per the standard.

- Severity- Evaluates the impact of the failure
- Occurrence – Evaluates the chance of a failure happening
- Detection -Evaluates the chance of a failure being detected

Risk priority number (RPN) = severity X occurrence X detection

Item & Function	Failure Mode	Effects of Failure	S	Causes of Failure	O	D	RPN	Recommended Actions	Actions Taken	S	O	D	RPN
Tankage Construction	Unable to engage optimum manpower/machineries	Time Delay	9	Sub Vendor Capability	8	3	216	Split the contracts into packages and Sub Contracts to be provided to two to three vendors	Contracts have been splitted into three packages and given to three vendors	3	2	5	42
				Insufficient Cash flow	9	5	405	Allow contractor to process the bills Bi-Weekly and ensure the Processing of payment on time	Contractor processed 46 Running Bills and Lead time of each bill shall not be exceeded more than 7 days	3	3	5	45
	Contractor Poor Performance	Time Delay	9	Sub Vendor Capability	7	10	630	Split the contracts into packages and Sub Contracts to be provided to two to three vendors	Contracts have been splitted into three packages and given to three vendors	7	6	10	420
				Insufficient Cash flow			630	Allow contractor to process the bills Bi-Weekly and ensure the Processing of payment on time	Contractor processed 46 Running Bills and Lead time of each bill shall not be exceeded more than 7 days				
	Labour leaving the site without any information	Time Delay	9	Improper payment	5	10	450	Labour payment to be ensured on monthly basis and Contractor to attach the payslip along with everybills	Labour payment was ensured on monthly basis. Contractor Running Bills were processed against the submission of Labour payment submission	7	4	8	280

Figure 17—PFMEA Table

Major Actions Taken from PFMEA:

1. Sub Contracts have been divided into three packages and given to three vendors
2. Sub Contracts have been given strategically to 3 different vendors that is 2 HSD Tanks & SKO Tanks given to one vendor and other 2 HSD tanks along with 4 MS tanks to a second vendor and remaining tanks were given to a third vendor.
3. Contractor and Labour payment had been ensured by Client and Consultant.

PFMEA Analysis for JACK operation:

Deck Lifting using Jack method activity was very critical so detailed PFMEA was done separately for Jack operation and all major issues were resolved before starting of the activity.

Item & Function	Failure Mode	Effects of Failure	S	Causes of Failure	O	Design Controls	D	RPN	Recommended Actions	Actions Taken	S	O	D	RPN
Hydraulic Jack operation	Structural Failure	can injure labour, damage plate	9	To much downward force, component failure	5	Ensure that the jack can be produced consistently with a high factor of safety	5	225	-	Detailed load calculation done and provided 6 stages of rings in Deck plate	3	2	5	42
	Rusting of jack over time	Failure of jack	9	Steel the material of the jack, can rust over time making it brittle	5	Make sure that coating is well applied	5	225	-	Steel coated to protect from elements, also placement during storage keep it protected from elements	3	3	5	45
	Uneven lifting of plates	can injure labour, damage plate	9	Useage of jack on uneven ground	7	Detection not possible from a pre-release standpoint	10	630	Study base more indepth to see how well it adapts for uneven ground, and study connection point with frame	Include more written and pictographic documentation with and printed on jack saying to use on level ground	7	6	10	420
	Labour gets fingers/hand caught in pinch point	Labour injury	9	Careless or uninformed labour	5	Detection not possible from a pre-release standpoint	10	360	Study pinch points	Include more written and pictographic documentation with, and printed on jack, cautioning about the pinch points	7	4	10	280
	Labour not using device as intended	Labour injury, damage to other items	9	Any number of reasons including jack slipping or applying too much force	2	Detection not possible from a pre-release standpoint	10	180	Think about how often this happens, labour misuse	Include more written and pictographic documentation with and printed on jack saying to use on the device as intended	7	2	10	140
	Labour scraping knuckles on ground	Labour injury	8	Position of jack and design of rod for turning	8	Detection not possible from a pre-release standpoint	10	640	Study deisgn and look at ways to improve this problem	Include more written and pictographic documentation with, and printed on jack, cautioning about the pinch points	7	6	10	420
	Labour having difficulty aligning with frame	Labour injury	8	Location of frame makes it difficult to spot where the jack needs to go in daylight as well as at night	7	Detection not possible from a pre-release standpoint	10	560	Study deisgn and look at ways to improve this problem	Include more written and pictographic documentation with, and printed on jack, cautioning about the pinch points	7	4	10	280

Figure 18—PFMEA Table for Jack Operation

Digitalization of Daily Progress Report

The Daily Progress Report (DPR) provides the management with a comprehensive overview of the project, including the current status, milestones achieved, mobilization of manpower and machinery, pending document approvals from consultants, escalations or issues, and other critical factors influencing project completion. Traditionally, DPRs were prepared manually, which involved a significant amount of administrative work for the officers approximately 30% of their effort, with nearly 250 entries per day making it a tedious and time-consuming task. Also, one of the major challenges faced during construction was inspection delays and timely document approvals. Initially, the daily progress and issues were recorded manually.

To streamline this process, we developed an online Microsoft User Form that enables contractors to update daily progress and issues directly through their mobile devices. The data entered is captured in a response sheet, and the HPCL reporting template automatically fetches this data using a Visual Basic program. This digital solution ensured that all issues were routed directly to the Project In-Charge, resulting in the elimination of inspection delays for 84 out of 127 days.

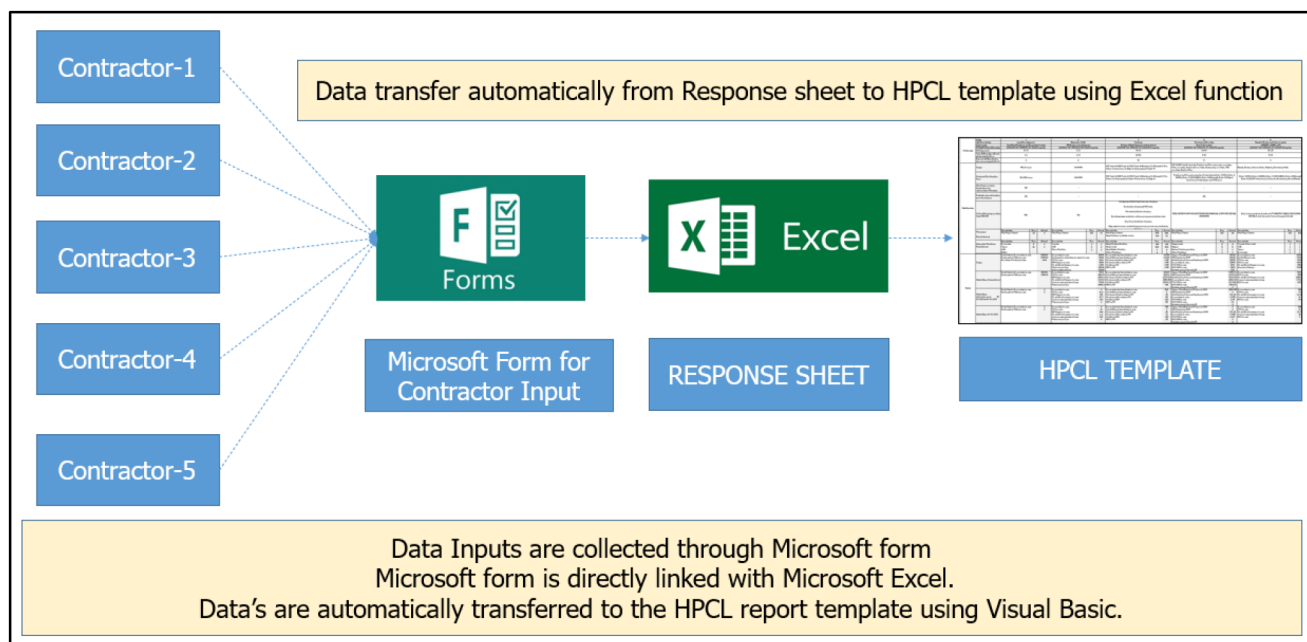


Figure 19—Process Flow Diagram

Validation Phase

Validation phase helps to validate the improvements suggested by the Cross Functional Team by PERT and Pareto Analysis after improvement

PERT Analysis after Improvement

PERT Analysis was carried out again to validate the project goal with given improvement. Three changes were made after improvement. Deck works time period reduced from 45 days to 7 days, Shell erection time reduced from 20 days to 18 days, Hydro testing time reduced from 30.33 days to 28.33 days.

Project lead time was reduced from 169 days to 127 days which is higher than the project goal.

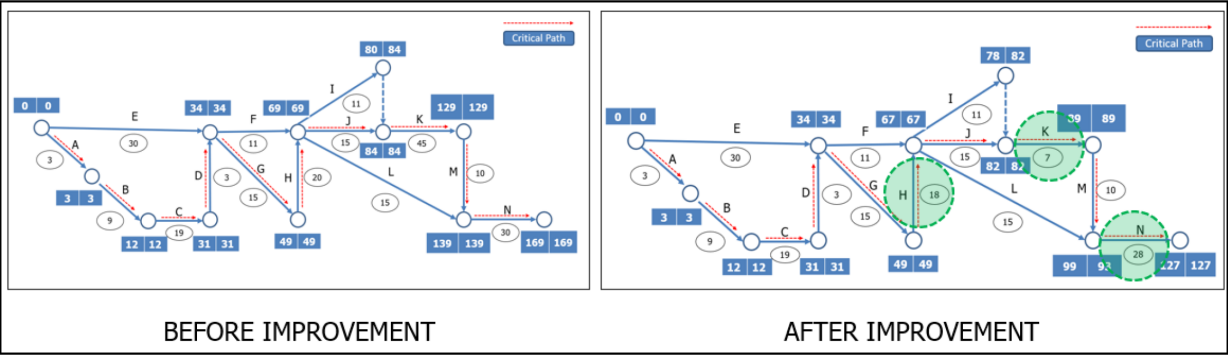


Figure 20—Network Diagram Before Vs After

Validation Phase

Productivity Improvement



Figure 21—Lead Time Improvement

Financial Improvement

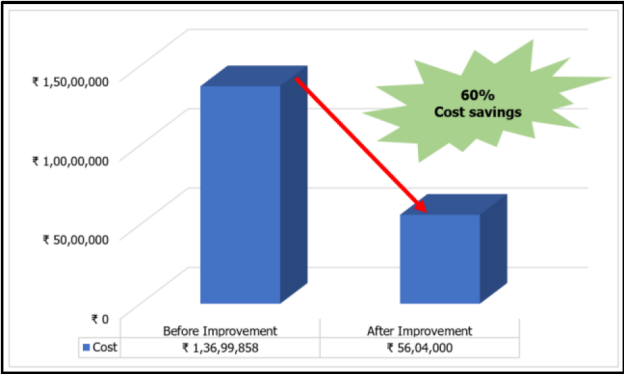


Figure 22—Financial Improvement

Intangible Benefits

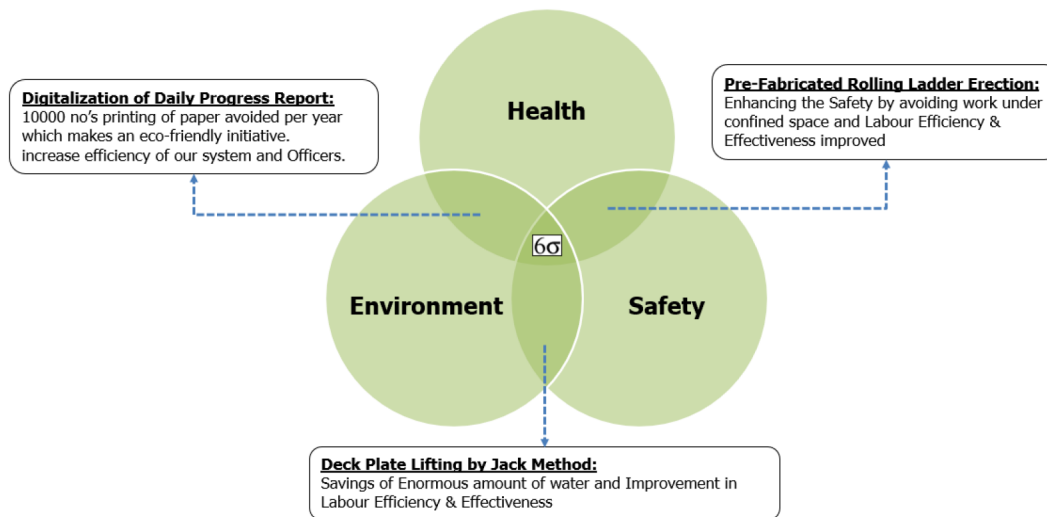


Figure 23—In-Tangible Benefits

Conclusion

In this DMAIC-based initiative, the tankage construction duration was reduced by 25%, yielding a cost saving of \$.137041 This performance improvement exemplifies the strength of the Six Sigma approach in project management. By using data-driven analysis to pinpoint inefficiencies and firmly address root causes, the DMAIC methodology not only cut timelines but also minimized waste and rework. These results align well with documented benefits of Six Sigma projects, which consistently deliver operational efficiency, cost reduction, and enhanced quality.

Abbreviations

DMAIC	Define Measure Analyze Improve Control
DMADV	Define Measure Analyze Design Validation
SMART	Specific Measurable Achievable Realistic and Time Bound
POL	Pipeline Oil Lubricants
CDD	Contractual Delivery Date
SIPOC	Supply Input Process Output Customer
PERT	Programme Evaluation and Review Technique
PFMEA	Potential Failure Modes and Effect Analysis
5W2H	What, Why, Where, When, Who, How, How much
MT	Metric Tonnage
KL	Kilo Liter
EPCM	Engineering, Procurement, and Construction Management

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